Agent Skills

Test selection, non-selection, and visible failure

Session 12 | 3 hours

Agenda

Segment	Time
Skill directory and loading model	8 min
Required frontmatter and description	10 min
Trigger and non-trigger design	12 min
Preconditions and visible failure	12 min
Validation and bounded runner	10 min
Candidate review and lab handoff	8 min

A skill has two jobs

1. **Select the right task.**
2. **Guide the work after selection.**

A good procedure with a broad description still creates noise. A precise description with no failure behavior still creates unsafe success claims.

Repository skill structure

```
.github/
└── skills/
    └── api-design/
        └── SKILL.md
```

The directory uses lowercase words and hyphens. The instruction file is named exactly `SKILL.md`.

Required frontmatter

```
---
name: api-design
description: Use for creating or reviewing REST routes under src/api/ when acceptance criteria, a reviewer, and a verified focused test command are available. Do not use for documentation, UI, deployment, or generated clients.
---
```

The `name` matches the directory. The `description` carries the trigger.

Test the positive trigger

```
Review GET /api/tasks/:id under src/api/.
Check identifier handling and the existing not-found response.
```

Required context:

- acceptance criteria;
- named paths;
- reviewer;
- verified focused test command.

Expected status: `applied`.

Test a non-trigger

```
Rewrite the installation section in README.md.
```

Expected:

```
{
  "status": "not_applicable",
  "skill": "api-design",
  "reason": "Task is outside REST route work under src/api/."
}
```

A bounded skill stays quiet outside its job.

Matching work can still be blocked

```
Add POST /api/tasks under src/api/.
Acceptance criteria exist.
No reviewer is assigned.
```

The trigger matches. The precondition does not.

Make the failure visible

```
{
  "status": "blocked",
  "skill": "api-design",
  "reason": "Missing precondition: human reviewer."
}
```

Do not invent a reviewer, skip the check, or return success.

A reviewable skill body

```
Preconditions
Procedure
Validation
Failure behavior
Maintenance
```

Each section answers a different question. Keep the procedure short enough that a human can apply it as a checklist.

Validate the files

```
npm run validate:skill
```

The validator checks:

- `SKILL.md` exists;
- frontmatter has `name` and `description`;
- `name` matches the directory;
- required sections are present.

Test all three outcomes

```
npm test
```

The suite proves:

- a route task applies the skill;
- a documentation task does not;
- a missing reviewer blocks;
- valid and invalid directories are distinguishable.

End with a bounded run

```
npm run demo
```

The output shows:

1. correct application to named API files;
2. a visible missing-reviewer failure.

The local runner is a teaching harness, not a replacement skill platform.

Review candidates before adaptation

Check:

- source and license;
- files and scripts;
- dependencies and network use;
- data flow;
- repository fit;
- owner and retirement condition.

Do not run an unfamiliar script just to inspect it.

Lab handoff

1. Create `api-design/SKILL.md`.
2. Validate the directory and frontmatter.
3. Run the trigger scenario.
4. Run the non-trigger scenario.
5. Run the missing-reviewer scenario.
6. Finish with the bounded demo.
7. Record live loading as executed or **not executed**.

The final evidence must show both correct application and a visible precondition failure.